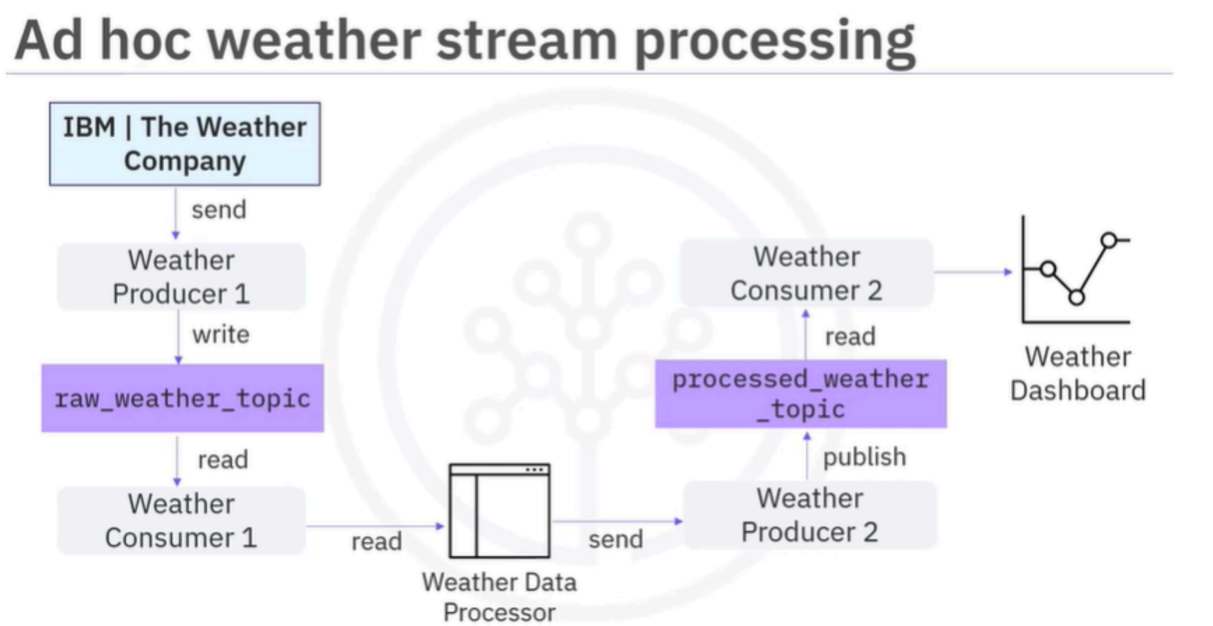
Kafka Streaming Process

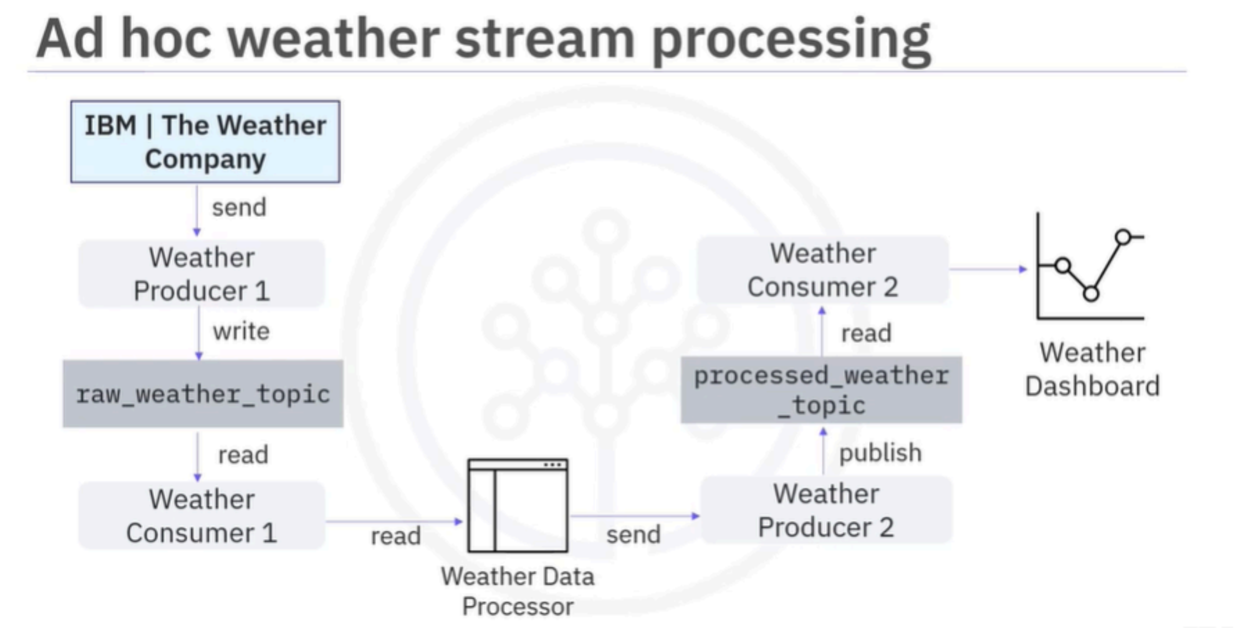
Sure! Let's talk about the Kafka Streams API, which is a powerful tool for processing data in real-time.

The Kafka Streams API is like a helpful assistant that takes in data from different sources, processes it, and then sends it out to where it needs to go. Imagine you have a busy kitchen where different ingredients (data) come in. The Kafka Streams API acts like a chef who takes those ingredients, prepares them (processes them), and serves the finished dish (processed data) to the dining area (another topic). This makes it easier to handle lots of data without getting overwhelmed!

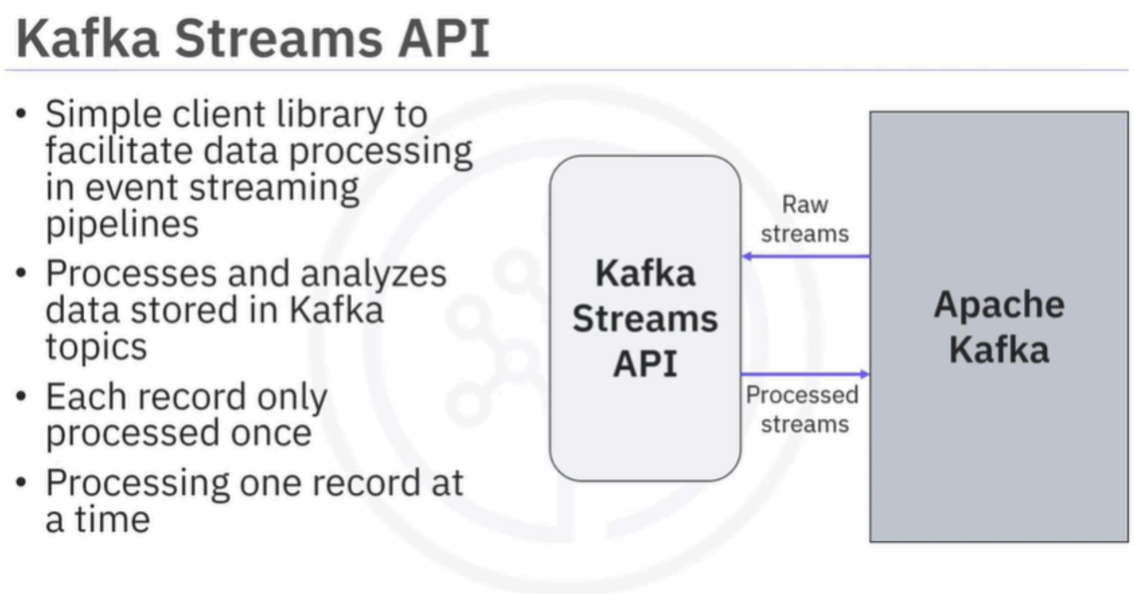


- In event streaming, in addition to transporting data, data engineers also need to process data through. For example, data filtering, aggregation, and enhancement. Any applications developed to process streams are called stream processing applications. For stream processing applications based on

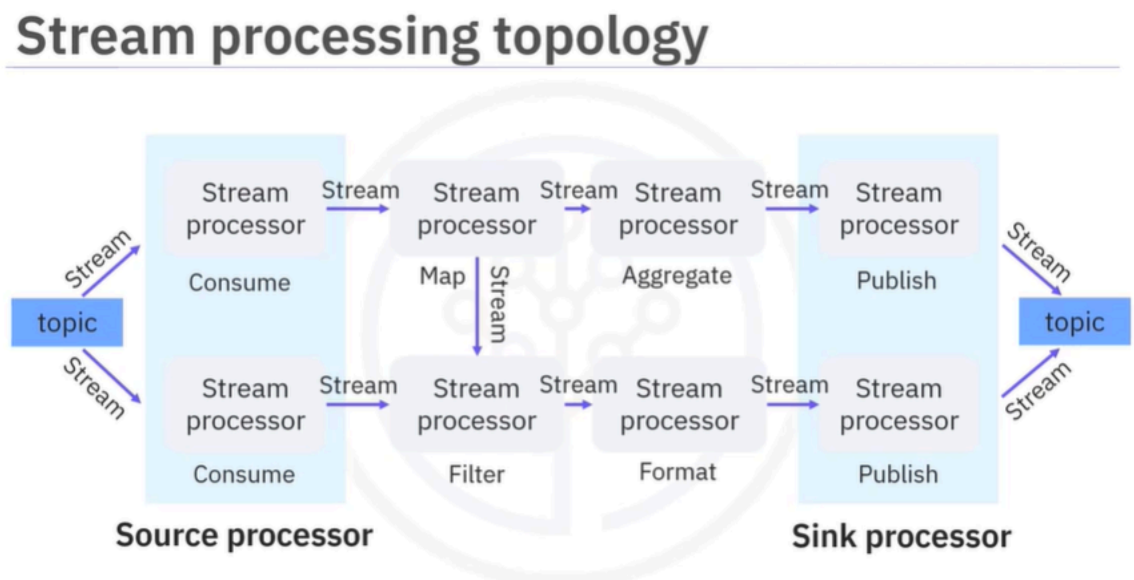
Kafka, a straightforward way is to implement an ad hoc data processor to read events from one topic, process them, and publish them to another topic.



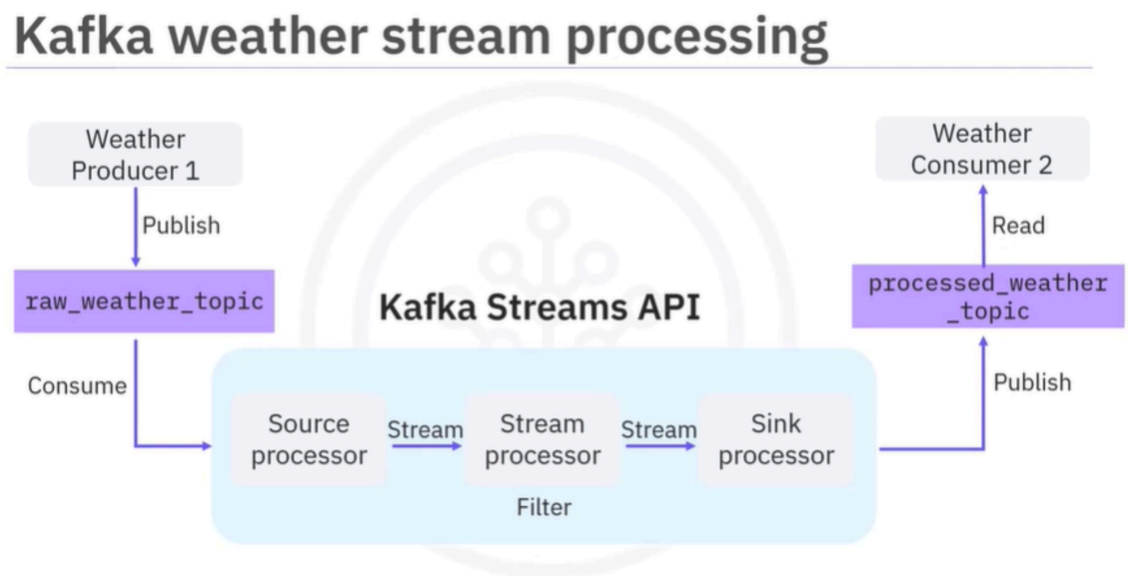
- Let's look at an example. You first request raw weather JSON data from a weather API, and you start a weather producer to publish the raw data into a raw_weather_topic. Then you start a consumer to read the raw weather data from the weather topic.
- Next, you create an ad hoc data processor to filter the raw weather data to only include extreme weather events, such as very high temperatures. Such a processor could be a simple script file or an application which works with Kafka clients to read and write data from Kafka. Afterwards, the processor sends the processed data to another producer and it gets published to a processed_weather_topic.
- Finally, the processed weather data will be consumed by a dedicated consumer and sent to a dashboard for visualization.



- Such ad hoc processors may become complicated if you have many different topics to be processed. A solution that may solve these challenges is Kafka. It provides the Streams API to facilitate stream processing. Kafka Streams API is a simple client library aiming to facilitate data processing in event streaming pipelines. It processes and analyzes data stored in Kafka topics. Thus, both the input and output of the Streams API are Kafka topics. Additionally, Kafka Streams API ensures that each record will only be processed once. Finally, it processes only one record at a time.



- Kafka Streams API is based on a computational graph called a stream processing topology. In this topology, each node is a stream processor, which receives streams from its upstream processor; performs data transformations, such as mapping, filtering, formatting, and aggregation; and produces output streams to its downstream stream processors.
- Thus, the edges of the graph are the I/O streams. There are two special types of processors. On the left, you can see the source processor which has no upstream processors. A source processor acts like a consumer, which consumes streams from Kafka topics and forwards the process streams to its downstream processors. On the right, you can see the sink processor, which has no downstream processors. A sink processor acts like a producer which publishes the received stream to a Kafka topic.



- Let's redesign the previous weather stream processing application with Kafka Streams API. Suppose you have a raw_weather_topic and a processed_weather_topic in Kafka. Now, instead of spending a huge amount of effort developing an ad hoc processor, you could just plug in the Kafka Streams API here.
- In the Kafka Streams topology, we have three stream processors, the source processor that consumes raw weather streams from the raw_weather_topic and forwards the weather stream to the stream processor to filter the stream based on high temperature.
- Then the filtered stream will be forwarded to the sink processor, which then publishes the output to the processed_weather_topic. Concluding, this is a much simpler design than an ad hoc data processor, especially if you have many different topics to be processed.

Recap

In this video, you learned that:

- Kafka Streams API is a simple client library supporting you with data processing in event streaming pipelines
- A stream processor receives, transforms, and forwards the processed stream
- Kafka Streams API uses a computational graph
- There are two special types of processors in the topology: The source processor and the sink processor